

Java String Operations

1. length()

```
public class StringLength {  
    public static void main(String[] args) {  
        String name = "Java";  
  
        // length() returns the number of characters in the String.  
        System.out.println("String: " + name);  
        System.out.println("Length: " + name.length());  
    }  
}
```

Output

Command Prompt

```
Microsoft Windows [Version 10.0.19045.4046]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\Student>javac StringLength.java  
C:\Users\Student>java StringLength  
String: Java  
Length: 4
```

2. charAt()

```
public class CharacterExample {  
    public static void main(String[] args) {  
        String name = "Java";  
  
        // charAt(index) returns the character at the given index.  
        // String indexing starts from 0.  
        System.out.println(name.charAt(0));  
        System.out.println(name.charAt(1));  
        System.out.println(name.charAt(2));  
        System.out.println(name.charAt(3));  
    }  
}
```

Output

Command Prompt

```
Microsoft Windows [Version 10.0.19045.4046]
(c) Microsoft Corporation. All rights reserved.
```

```
C:\Users\Student>javac CharacterExample.java
```

```
C:\Users\Student>java CharacterExample
```

```
J
a
v
a
```

3. concat()

```
public class StringConcat {
    public static void main(String[] args) {
        String first = "Hello";
        String second = "World";

        // concat() joins the second String to the first String.
        System.out.println(first.concat(" ").concat(second));
    }
}
```

Output

Command Prompt

```
Microsoft Windows [Version 10.0.19045.4046]
(c) Microsoft Corporation. All rights reserved.
```

```
C:\Users\Student>javac StringConcat.java
```

```
C:\Users\Student>java StringConcat
```

```
Hello World
```

4. equalsIgnoreCase()

```
public class EqualsIgnoreCaseExample {
    public static void main(String[] args) {
        String s1 = "JAVA";
        String s2 = "java";

        // equalsIgnoreCase() compares Strings without considering case.
        System.out.println(s1.equalsIgnoreCase(s2));
    }
}
```

Output

Command Prompt

```
Microsoft Windows [Version 10.0.19045.4046]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Student>javac EqualsIgnoreCaseExample.java
C:\Users\Student>java EqualsIgnoreCaseExample
true
```

5. toUpperCase()

```
public class UpperCaseExample {
    public static void main(String[] args) {
        String name = "Java Programming";

        // toUpperCase() converts all letters to uppercase.
        System.out.println(name.toUpperCase());
    }
}
```

Output

Command Prompt

```
Microsoft Windows [Version 10.0.19045.4046]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Student>javac UpperCaseExample.java
C:\Users\Student>java UpperCaseExample
JAVA PROGRAMMING
```

6. toLowerCase()

```
public class LowerCaseExample {
    public static void main(String[] args) {
        String name = "JAVA PROGRAMMING";

        // toLowerCase() converts all letters to lowercase.
        System.out.println(name.toLowerCase());
    }
}
```

Output

Command Prompt

```
Microsoft Windows [Version 10.0.19045.4046]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Student>javac LowerCaseExample.java
C:\Users\Student>java LowerCaseExample
java programming
```

7. substring()

```
public class SubstringExample {
    public static void main(String[] args) {
        String name = "Programming";

        // substring(0, 4) starts at index 0 and stops before index 4.
        System.out.println(name.substring(0, 4));

        // substring(4) returns the String from index 4 to the end.
        System.out.println(name.substring(4));
    }
}
```

Output

Command Prompt

```
Microsoft Windows [Version 10.0.19045.4046]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Student>javac SubstringExample.java
C:\Users\Student>java SubstringExample
Prog
ramming
```

8. indexOf()

```
public class IndexExample {
    public static void main(String[] args) {
        String name = "Java Programming";

        // indexOf() returns the position of the first matching character.
        System.out.println(name.indexOf('a'));

        // It can also find the starting position of a String.
        System.out.println(name.indexOf("Programming"));
    }
}
```

Output

Command Prompt

```
Microsoft Windows [Version 10.0.19045.4046]  
(c) Microsoft Corporation. All rights reserved.
```

```
C:\Users\Student>javac IndexExample.java
```

```
C:\Users\Student>java IndexExample
```

```
1
```

```
5
```

9. lastIndexOf()

```
public class LastIndexExample {  
    public static void main(String[] args) {  
        String name = "banana";  
  
        // indexOf() finds the first occurrence of 'a'.  
        System.out.println(name.indexOf('a'));  
  
        // lastIndexOf() finds the last occurrence of 'a'.  
        System.out.println(name.lastIndexOf('a'));  
    }  
}
```

Output

Command Prompt

```
Microsoft Windows [Version 10.0.19045.4046]  
(c) Microsoft Corporation. All rights reserved.
```

```
C:\Users\Student>javac LastIndexExample.java
```

```
C:\Users\Student>java LastIndexExample
```

```
1
```

```
5
```

10. replace()

```
public class ReplaceExample {  
    public static void main(String[] args) {  
        String name = "Java";  
  
        // replace() replaces every matching character.  
        System.out.println(name.replace('a', 'o'));  
    }  
}
```

Output

Command Prompt

```
Microsoft Windows [Version 10.0.19045.4046]  
(c) Microsoft Corporation. All rights reserved.
```

```
C:\Users\Student>javac ReplaceExample.java  
C:\Users\Student>java ReplaceExample  
Jovo
```

11. contains()

```
public class ContainsExample {  
    public static void main(String[] args) {  
        String name = "Java Programming";  
  
        // contains() returns true when the given text exists in the String.  
        System.out.println(name.contains("Java"));  
        System.out.println(name.contains("Python"));  
    }  
}
```

Output

Command Prompt

```
Microsoft Windows [Version 10.0.19045.4046]  
(c) Microsoft Corporation. All rights reserved.
```

```
C:\Users\Student>javac ContainsExample.java  
C:\Users\Student>java ContainsExample  
true  
false
```

12. trim()

```
public class TrimExample {  
    public static void main(String[] args) {  
        String name = "  Java  ";  
  
        // trim() removes leading and trailing spaces.  
        System.out.println("Before: " + name);  
        System.out.println("After: " + name.trim());  
    }  
}
```

Output

Command Prompt

```
Microsoft Windows [Version 10.0.19045.4046]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Student>javac TrimExample.java
C:\Users\Student>java TrimExample
Before:    Java
After: Java
```

13. isEmpty()

```
public class EmptyExample {
    public static void main(String[] args) {
        String s1 = "";
        String s2 = "Java";

        // isEmpty() returns true only when the String has zero characters.
        System.out.println(s1.isEmpty());
        System.out.println(s2.isEmpty());
    }
}
```

Output

Command Prompt

```
Microsoft Windows [Version 10.0.19045.4046]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Student>javac EmptyExample.java
C:\Users\Student>java EmptyExample
true
false
```

14. compareTo()

```
public class CompareToExample {
    public static void main(String[] args) {
        String s1 = "Apple";
        String s2 = "Banana";

        // compareTo() compares Strings lexicographically.
        // Negative means s1 comes before s2.
        // Zero means both Strings are equal.
        // Positive means s1 comes after s2.
        System.out.println(s1.compareTo(s2));
        System.out.println(s1.compareTo("Apple"));
        System.out.println(s2.compareTo(s1));
    }
}
```

Output

Command Prompt

```
Microsoft Windows [Version 10.0.19045.4046]
(c) Microsoft Corporation. All rights reserved.
```

```
C:\Users\Student>javac CompareToExample.java
```

```
C:\Users\Student>java CompareToExample
```

```
-1
```

```
0
```

```
1
```

15. toCharArray()

```
public class CharArrayExample {
    public static void main(String[] args) {
        String name = "Java";

        // toCharArray() converts the String into a character array.
        char[] arr = name.toCharArray();

        // The enhanced for loop prints each character from the array.
        for (char c : arr) {
            System.out.println(c);
        }
    }
}
```

Output

Command Prompt

```
Microsoft Windows [Version 10.0.19045.4046]
(c) Microsoft Corporation. All rights reserved.
```

```
C:\Users\Student>javac CharArrayExample.java
```

```
C:\Users\Student>java CharArrayExample
```

```
J
```

```
a
```

```
v
```

```
a
```